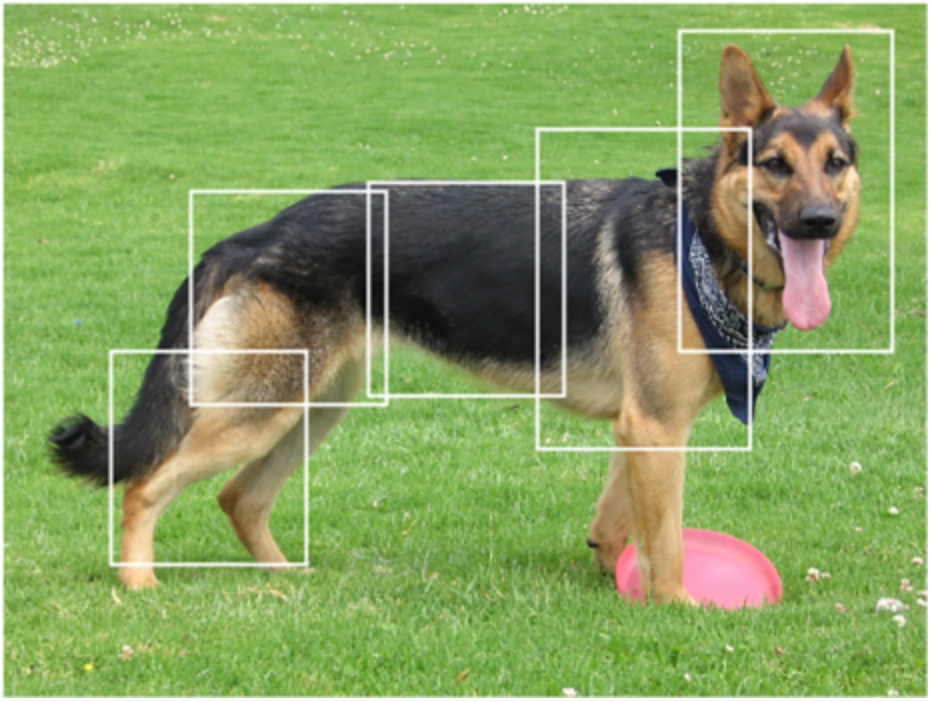


We could train such a model using a genetic algorithm, as you learned in chapter 6, or we could use reinforcement learning. In the reinforcement learning case, the model would produce an integer set of coordinates, crop the image based on those coordinates, process the subregion, and make a classification decision. If it classifies correctly, it would get a positive reward, or it would get a negative reward for an incorrect classification. In this way, we could employ the REINFORCE algorithm you learned earlier to train the model to perform an otherwise nondifferentiable function. This procedure is described in the paper "Recurrent Models of Visual Attention" by Volodymyr Mnih et al. (2014). This form of attention is termed *hard* attention because it is nondifferentiable.

There is also *soft* attention, which is a differentiable form of attention that simply applies a filter to minimize or maintain certain pixels in the image by multiplying each pixel in the image by a soft attention value between 0 and 1. The attention model can then learn to set certain pixels to 0 or maintain certain relevant pixels (figure 10.5). Since the attention values are real numbers and not integers, this form of attention is differentiable, but it loses the efficiency of a hard attention model, since it still needs to process the entire image rather than just a portion of it.